

Agent-Controlled Forgetting for Tool-Using Agents: Reversible Context Curation in Practice

Jan-Peter Franke
franke@bold-projekte.com

Münster, Germany · 22 September 2026

Abstract

Tool-using agents repeatedly carry observations whose useful content can be much smaller than their original payload. We study agent-controlled forgetting: the acting model selects previously observed tool results, replaces each with a short note at its original position, and retains the exact original in a recoverable archive. A Python harness exposes batch archival and explicit recovery without task-specific model training, while protecting user instructions and assistant messages from these operations. In an exploratory OpenTelemetry debugging case followed by an unrelated implementation task, the method ended with 231,951 provider-reported prompt tokens versus 912,492 under retained history, used 50% fewer cumulative input tokens, and had an estimated API cost of \$1.28–\$1.44 versus approximately \$4.38. Both arms passed the two-case primary behavioral oracle; neither fully satisfied the follow-up evaluation. The method made more requests and took 17% longer. A contrasting application-development pair produced no context or cost saving, and an earlier continuation exhibited lower manually assessed quality despite reduced context. These observations demonstrate substantial resource savings in noisy tool-use trajectories and identify workload dependence as a central consideration for reversible context management.

1 Introduction

A debugging agent may inspect an entire distributed trace to identify one failed service call. A coding agent may need the final diagnostic from a long build log, while the next request still contains every preceding progress line. These observations can be useful to inspect once without being useful to retain in full throughout the remaining task. The resulting research question is not only how much information an agent can receive, but which already inspected information should continue to occupy its active context.

The design draws on human practices of organizing working material while using a software-specific capability: explicitly setting an observation aside and later restoring its exact contents. An organized desk provides the analogy: keep current evidence within reach and file completed material with a short note and a retrieval reference.

We investigate a separation between **exposure** and **continued retention**. After observing a tool result, the same model that performs the task can decide whether to keep that result verbatim or replace it with an explicit note. The original remains recoverable. This decision can use knowledge gained during diagnosis, including the fact that a previously plausible cause has been ruled out or that the user has moved to a different task. The mechanism does not avoid the initial cost of reading a result. Its potential benefit lies in reducing repeated input and leaving less obsolete material in subsequent requests.

This distinction matters when the agent cannot change the tools that produce its observations. A third-party Model Context Protocol (MCP) server may return a broad response; a browser

observation may include navigation and controls unrelated to the current question; a test runner may emit extensive logs before a short verdict. Improving those interfaces or filtering their outputs can reduce initial input. However, selection before the acting model sees a result may omit unexpected evidence, and some material becomes dispensable only after the model interprets it. Post-exposure curation gives the agent a further choice over continued retention without requiring changes to the source tool.

The mechanism originates in the author’s Arios agent, with extraction, recovery, and explicit forgetting tools recorded in its local implementation history in February 2026; the companion development evidence documents this history.

Long-context evaluation has shown that the position of relevant information can affect retrieval and question-answering performance in studied models (Liu et al. 2024). This motivates measuring context use, but does not imply a universal token threshold at which an agent becomes unreliable. A smaller prompt is also not a measurement of greater reasoning ability. We therefore report active prompt size, cumulative usage, estimated cost, and task outcomes separately.

Agent-managed memory is an established research direction. MemGPT provides model-directed movement between memory tiers; MemAct integrates context editing into the acting policy; Context-Folding structures temporary subtrajectories; ACM provides agent-initiated summarization and archive queries; and AdaCoM trains a separate context manager. Our contribution is an implementation and exploratory evaluation of a restricted, auditable operation over individual tool-result occurrences, not a claim to invent agent-controlled memory. We study how this operation behaves in real coding and browser trajectories, including a case in which retained history exceeds 900,000 reported prompt tokens.

The report contributes: (1) a concrete reversible tool-result contract with stable identifiers, in-place notes, protected instruction messages, and atomic batch validation; (2) an independently graded debugging-and-pivot case with provider usage and cache-aware cost accounting; and (3) contrasting and adverse observations that delimit where the mechanism is useful. The central case demonstrates substantial reductions in active context and repeated input while preserving success on the tested primary behaviors.

2 Related work

Memory tiers and agent-directed editing. Packer et al. (2023) introduce MemGPT, where function calls move information between active and external memory and recall storage preserves earlier interactions. Its queue manager also controls overflow. Our evaluated operation targets selected tool outputs in the current conversation, retaining a note at each original location. This is a narrower interface choice within the broader area of agent-managed memory.

Learned management policies. Zhang et al. (2026) formulate MemAct around a Prune&Write operation over identifiable interaction records and train the policy using Dynamic Context Policy Optimization. Thus, allowing the task agent to choose context edits is already established. Yi et al. (2026) instead train an external manager that rewrites, deletes, or merges messages for a frozen acting agent. Our experiments use a prompted acting model without additional training or a separate manager. They do not compare against either learned system.

Structured reduction and accumulated knowledge. Sun et al. (2025) introduce Context-Folding, which branches into subtrajectories and returns a summary while removing intermediate context, with FoldGRPO training. Our operation can select already completed results without requiring a branch boundary. Zhang et al. (2025) describe ACE, where generation, reflection, and curation update an evolving playbook. Maintaining reusable knowledge and reducing active observation payload are complementary goals; we do not evaluate that combination here.

Proactive folding and recoverable transcripts. Ye et al. (2025) train AgentFold to combine folding directives with web-agent actions, maintaining summaries at multiple scales alongside the latest interaction. This is another precedent for treating context management as part of agent behavior. The practical pi-fold implementation (Conner, n.d.) preserves exact transcript spans outside the active window, replaces them with briefs, and supports expansion. Its inspected README distinguishes earlier agent-laid marks from a newer automatic folding policy. Our interface targets individual tool-result occurrences selected by the acting model and returns recovered payloads as new results; the shared use of briefs, external originals, and recovery is not a novelty claim. We have not benchmarked against pi-fold.

Retrieval and persistent-memory retention. Nahid and Rafiei (2025) introduce PRISM, whose Selector and Adder iteratively refine an evidence set for multi-hop question answering. This demonstrates why filtering should not be reduced to a single irreversible preprocessing step. Wei et al. (2026) describe FadeMem’s adaptive decay, memory layers, conflict resolution, and fusion. Rusu et al. (2026) extract conversational content into a graph and prune nodes using retention scores. These systems manage retrieval candidates or persistent memory representations, whereas our measured operation changes the continued presence of raw observations in an active trajectory.

Compression policies and archival. Kang et al. (2025) introduce ACON, which optimizes compression guidelines from trajectory feedback and uses a separate model to compress histories or observations at token thresholds. Our interface instead lets the acting model retrospectively select individual tool results and author their replacement notes. Li et al. (2026) introduce ACM, which shares agent-initiated management and archival of raw messages, but summarizes message intervals with a separate LLM and retrieves query-relevant extracts through another model. Our contract preserves user, system and assistant messages and returns exact archived payloads on recovery. ACM evaluates both an untrained management framework and a post-trained policy; absence of task-specific training alone is therefore not a distinguishing feature.

Provider-side editing. Anthropic’s context-editing documentation (Anthropic, n.d.) describes threshold-triggered removal of older tool results with placeholders, while the client retains the original conversation. It also documents cache invalidation when results are cleared. This provides a practical comparison point for rule-based editing. Our model-selected notes and explicit recovery interface are choices within this broader design space; no provider-side baseline is evaluated here.

Positioning. Pre-exposure retrieval and filtering can reduce what the agent initially receives. Post-exposure curation can additionally exploit conclusions reached after inspection. This timing distinction does not imply that filters are intrinsically inferior: iterative retrieval, query refinement, and programmatic processing can also adapt to task state. Our retained-history comparator isolates a practical policy bundle, not the best available context-management baseline. The targeted literature review does not establish novelty of the full combination of archival, notes, identifiers, and recovery.

3 Method

3.1 State and eligible operations

Let the active conversation be an ordered message sequence H , with an external archive A . An eligible tool-result occurrence contains an identifier r , tool-call metadata, text x , and optional image payloads v . Stable identifiers are assigned when results enter the conversation. They start at r_1 in a new session and are never reused within that session. The result occurrence, rather than its source file or tool name, is the unit of selection.

The acting model calls `context_apply` with a batch of pairs (`result_id`, `note`). For every selected result, the harness stores its raw text, images, and provenance in A, then replaces the active result body with a stub containing the note and archive reference. The associated tool call remains in the trajectory. A note may retain an extract, a diagnostic conclusion, source locations, or a reason why the result is no longer needed.

`context_recover(archive_ref)` returns the original as a new tool result at the current end of the conversation. The recovered occurrence receives a new result identifier. Recovery does not rerun the original tool: an archived browser page or test log remains evidence of the earlier observation, not a fresh environmental measurement. Original stubs remain addressable.

The active representation is lossy because a note need not preserve every detail. Reversibility refers to retrieval of the stored payload, not to lossless reasoning or preservation of every future decision dependency. Likewise, forgetting here is eviction from the active request, not deletion from storage, model unlearning, or a privacy-erasure mechanism.

3.2 Integrity and protocol boundaries

The backend validates the entire batch before committing a change. Unknown, duplicate, already archived, or otherwise unavailable identifiers produce a corrective error without partial mutation. Originals are archived before stubs are committed. User, system, and assistant messages are not selectable through the context tools. Archival does not edit workspace files.

A compact description of the transition is:

```
apply(batch):
    resolve every result ID and validate every note
    reject the whole batch if any target is invalid
    stage originals and provenance in the archive
    commit notes and archive references at the selected result locations
    return the result-ID to archive-reference mapping

recover(ref):
    load the stored text and image payloads
    append them as a new tool result with a fresh result ID
```

The trace preserves original observations independently of the active representation. This enables comparison of archived payloads against recorded tool returns. Such equality checks validate storage integrity; they do not validate the semantic adequacy of notes. The provider projection also handles API representation constraints, so preservation of stored assistant responses should not be confused with byte-identical wire serialization in every error case.

3.3 Prompted selection

The method arm receives context-management instructions, tool descriptions, examples, and an ephemeral runtime reminder. The acting model is asked to curate after useful research or diagnostic progress, batch substantial targets, retain unresolved evidence, and continue the original task. The prompt mentions roughly 25% irrelevant content as a heuristic, together with material size reduction and call overhead. This percentage is neither an annotated dataset property nor an enforced classifier threshold.

The final runtime implementation estimates current text size from the projected request and makes a reminder eligible at approximately 40,000 text tokens, or when image content is present. After a reminder, another requires meaningful new eligible output, configured as 20,000 characters or a new image. A new user task resets reminder eligibility. The reminder is appended as

user-role content at the end of the outgoing request and is not persisted in conversation history. It describes maintenance rather than a deadline or an instruction to finish.

The condition therefore tests **prompted, model-selected curation**. It does not show that an unprompted base model spontaneously discovers a need to forget, and it does not involve parameter learning during the experiment. Retained history omits the method instructions, reminder, and context tools while keeping the shared task tools and result labels.

3.4 Resource accounting

For request t , let P_t be provider-reported prompt tokens and O_t completion tokens. We report the last measured P_t , maximum P_t , cumulative input $\sum_t P_t$, and total reported tokens $\sum_t (P_t + O_t)$. The last prompt measurement precedes generation of the final submission; it is not an exact tokenization of the post-answer history. Prompt size includes the request representation and is an operational proxy for active context, not a direct measure of usable reasoning capacity.

If K_t cached input tokens are reported, estimated request cost is $r_{\text{input}}(P_t - K_t) + r_{\text{cache}}K_t + r_{\text{output}}O_t$. Missing cache detail produces a range rather than an assumed zero. Reasoning tokens reported within completion usage are not added again. For OpenTelemetry, recorded rates were \$0.20 per million ordinary input tokens, \$0.05 per million cached input tokens, and \$0.50 per million output tokens. These are run-start tariff estimates, not invoices or measurements of provider compute.

Curation may invalidate a reusable prompt prefix and increase the cost of the next request. A smaller subsequent history may nevertheless reduce total paid input. We measure the resulting usage and tariff estimate, without assuming either perfect cache preservation or complete cache loss. Wall time includes tool activity; it cannot identify a causal time-to-first-token effect.

4 Experimental design

4.1 Scope and development history

This is an exploratory systems study using the provider model identifier `z-ai/glm-5.3-flash` through TensorX. The principal comparisons use one fresh session per condition, not a random sample of independent tasks. Prompt and harness development preceded these comparisons. Earlier attempts, including unfavorable outcomes and infrastructure failures, remain recorded. Selecting the high-noise case after earlier experiments makes it a development-informed case study rather than a held-out confirmatory benchmark.

We report the repaired OpenTelemetry pair as the central case, the completed Planroom pair as a workload contrast, and the earlier six-session Sphinx batch as descriptive supporting evidence. An adverse browser continuation is included explicitly. The experiments are not pooled into an overall accuracy score: tasks, graders, histories, and harness versions differ.

4.2 OpenTelemetry debugging and task pivot

The environment uses the public OpenTelemetry Demo source before an upstream checkout fix, at commit `af3dc6553d9f7aebb5929b192fbf3a922a467050`. The primary task is to diagnose a failing checkout using browser and observability surfaces, implement a safe frontend error contract, and verify it. Background load and concurrent payment, advertising, image, Kafka, home-page, and product-catalog faults produce realistic competing observations. No prose padding or mandatory reading volume is injected. The fraction of irrelevant material is not independently annotated.

After primary submission, the same session receives an unrelated C++ currency-service task requiring graceful shutdown. Each condition carries its own actual primary history and workspace into this phase. Consequently, the follow-up compares full trajectories, not an identical-history intervention that isolates the effect of the pivot itself.

Retained runs first and method second. Each starts from a reset stack and a byte-identical source snapshot. Shared model configuration, task contracts, application setup, and task tools are held fixed; curation instructions and tools differ by design. Live telemetry, browsing choices, and generated code are not identical across conditions. The runner configures 1,500 logical calls and 1,800 provider attempts per episode, with a 131,072-token output budget and requested low reasoning effort. Neither condition reaches those limits. No fixed provider sampling seed or explicit temperature is supplied in this runner.

The private primary oracle rebuilds a copy of the candidate and runs two upstream-derived Cypress cases. Both baseline failure and gold success were checked. Passing those cases supports the tested error behavior, not exhaustive satisfaction of every prose requirement.

The currency grader combines compilation, process probes, and source-text checks. It tests SIGTERM and SIGINT, process identity, exit status, elapsed time, and an occupied-port startup failure. Its six-criterion summary additionally uses string and regular-expression checks for implementation structure and reporting. The baseline scores 0/6 and the reference 6/6. However, calibration on one reference cannot establish general grader validity: alternative correct implementations may fail static patterns, and matching a pattern does not prove semantics. We therefore emphasize executable observations and treat the aggregate follow-up score as provisional supporting evidence.

Thirty-seven benchmark inputs and a 605-file snapshot were frozen before the final pair. Version 01 had a compromised follow-up build surface and an incompletely specified primary error contract. A retained-only version-02 development episode exposed further deadline-contract ambiguity. Those trajectories are retained but excluded from the version-03 comparison. No post-grading repairs or selective reruns were applied to its candidates.

4.3 Contrasting tasks

Planroom. A multi-phase application-development task requires library-documentation research through Context7, implementation, and twelve prescribed browser journeys through Playwright. Both arms use the same specification and scaffold, with a completion gate binding criterion identifiers to their full requirement text. Journey coverage is assessed from traces and independent checks. Model-written test counts differ and are not comparable accuracy denominators. File-picker behavior and deliberate persistence-failure injection remain evaluation gaps.

Sphinx publication manifest. Six fresh sessions receive three successive implementation requests on Sphinx 8.2.3. The predefined order is retained, method, method, retained, retained, method. Private graders contain 11, 20, and 24 cases across phases. This older experiment uses a 16,384-token output limit and a global 300-request guard. Its different configuration prevents treatment as a replication of the final OpenTelemetry interface. Cases within each grader share requirements and are not independent statistical samples.

5 Results

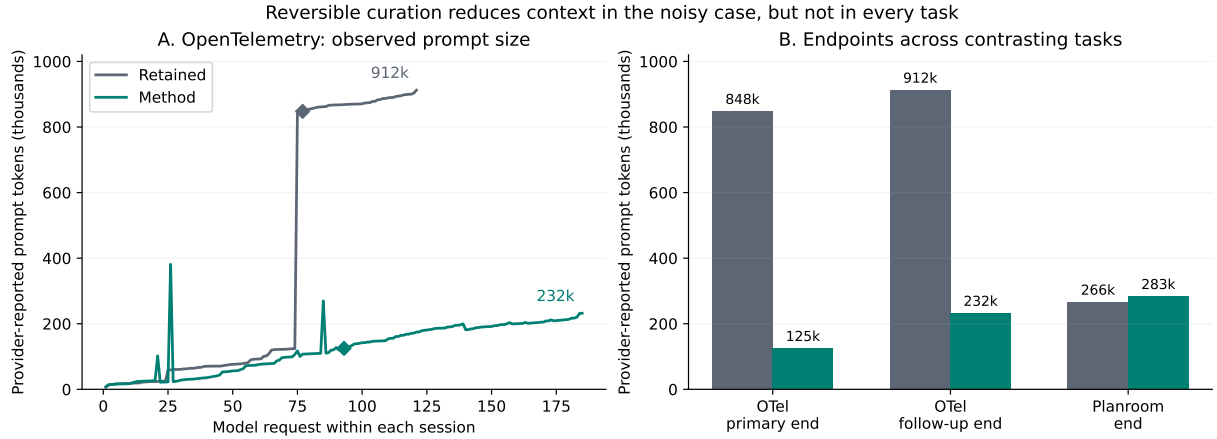


Figure 1. Provider-reported prompt size. Panel A plots each OpenTelemetry session against its own request index; diamonds mark primary submission, so equal horizontal positions do not denote equal work. Panel B includes the contrasting Planroom result. These are individual trajectories, with no uncertainty bands. The plotted endpoint is the input to the final response, not a post-answer context measurement.

5.1 OpenTelemetry resource use

Both arms submit both tasks normally, without recorded provider errors. Table 1 shows substantial reduction in active and cumulative input under the method despite more requests. At primary submission the method carries 124,746 prompt tokens versus 847,674 for retained, an 85% reduction. After the follow-up, the method carries 231,951 versus 912,492 prompt tokens, a 75% reduction. The method peak is 381,299 versus 912,492 tokens. Thus, the observed comparison reaches a substantially larger retained context than the earlier coding tasks.

Table 1. OpenTelemetry, both phases unless specified.

Metric	Retained	Method
Primary behavioral oracle	2/2	2/2
Primary submission prompt tokens	847,674	124,746
Final measured prompt tokens	912,492	231,951
Peak prompt tokens	912,492	381,299
Cumulative input tokens	45,766,146	22,988,240
Completion tokens	33,834	71,919
Input plus completion tokens	45,799,980	23,060,159
Model requests	121	185
Elapsed seconds	1,730	2,025
Estimated API cost, USD	4.378–4.379	1.283–1.436
Successful archival batches	0	9

Cumulative input decreases by 50%, and input plus completion tokens decrease by 50%. The estimated cost is approximately 67–71% lower, even after reported cache usage is included. The method generates more than twice as many completion tokens and takes 17% longer. These are resource savings without a speed advantage in this case. Original ledgers and exact deltas are linked in the OpenTelemetry metrics.

The largest retained-context jump follows a 2,478,818-character `browser_find` result at primary

step 74, three requests before primary submission; the follow-up then carries the accumulated history through 44 further requests. Because the arms encounter different observations and most primary implementation precedes this jump, the pair measures complete retention trajectories rather than controlled exposure to identical noise.

5.2 Observed task quality

Both primary candidates pass the two-case behavioral oracle.

For the follow-up, both candidates compile. The method exits with status 0 after SIGTERM in 8.43 seconds and after SIGINT in 8.85 seconds. Retained fails both probes and is killed with status 137. Conversely, retained passes the occupied-port startup-failure probe with status 1; the method’s second process does not finish within 30 seconds. The method therefore shows stronger tested signal handling but weaker tested startup-failure handling.

The frozen mixed grader reports 3/6 for method and 2/6 for retained. Source inspection identifies false negatives in its static checks: method calls `server_ptr->Shutdown(...)` rather than the expected `server->Shutdown(...)`, and retained uses `signalfd` rather than the expected `sigwait`. Both candidates contain remaining-budget calculations. Method’s startup-failure branch joins a signal-waiting thread without first arranging its termination, consistent with the observed timeout. The frozen grades are preserved, but their mixture of process probes and syntax-sensitive checks makes the individual runtime outcomes more informative than the aggregate score. See the retained and method records.

5.3 What was archived

The method makes five successful archival batches in the primary phase and four in the follow-up. In total, 26 results containing 2,247,229 text characters are represented by 9,709 note characters. Archived material includes large observability responses and completed diagnostic or verification output. All 26 archived text payloads and image lists match their recorded tool returns exactly. No recovery is used in this pair, and no image payload is archived.

The archive verifies exact payload preservation while the agent repeatedly curates context and continues working; this pair does not test retrieval behavior because no recovery is requested.

5.4 Workload contrast and repeated pilot

Table 2. Planroom completed pair.

Metric	Retained	Method
Supported browser journeys	12/12	12/12
Final and peak prompt tokens	265,965	283,105
Input plus completion tokens	67,669,624	68,136,934
Model requests	442	427
Elapsed minutes	31.78	24.29
Estimated API cost, USD	3.461–3.465	3.510–3.577

Planroom yields no context or cost saving: the method ends with 6% more prompt tokens (283,105 versus 265,965) and uses less than 1% more total tokens (68,136,934 versus 67,669,624). Both arms support the twelve journeys at the available assessment resolution, with the gaps described above. Archival of 34 results removes only 69,685 net history characters; this does not offset the differing generated trajectories and method overhead. The method completes in 24.29

rather than 31.78 minutes; this single pair does not isolate the cause of the timing difference. Planroom evidence.

Table 3. *Sphinx pilot, all six sessions in execution order.*

Session	Condition	Final private cases	Final prompt	Total tokens	Requests
01	Retained	23/24	267,349	15,946,799	116
02	Method	24/24	89,855	8,286,503	152
03	Method	24/24	145,213	16,849,371	209
04	Retained	24/24	268,937	20,217,639	136
05	Retained	23/24	234,057	12,273,223	98
06	Method	23/24	123,258	12,995,595	192

The Sphinx batch contains two final 24/24 method outcomes and one retained 24/24 outcome. Every method session has a smaller final prompt than every retained session, while cumulative usage overlaps. These six sessions characterize variation on one authored task under the older configuration, rather than general non-inferiority. Sphinx metrics.

5.5 Adverse and compromised observations

An earlier browser-based workflow-design continuation used 63% fewer total tokens with the method but received a manual score of 4/12 versus 6/12 for retained. Both answers had substantive state-machine errors. An explicitly requested method-only repetition later scored 6/12. That repeat was selected after observing the weak answer and must not replace it or count as a balanced replication. The original result therefore remains an observed quality loss alongside a resource saving. Adverse result, selected repeat.

Earlier development also encountered step caps, truncated output, invalid historical tool arguments, provider failures, and a faulty benchmark build surface. These failures explain subsequent harness changes and are not evidence of model degradation caused by context size. They remain part of the experiment history and are summarized in the companion evidence map.

6 Discussion

Selection after diagnosis. Large tool outputs can be necessary for discovering an explanation but redundant after that explanation is established. The OpenTelemetry archive illustrates that the agent can act on this distinction. Repeated input savings arise only after the initial exposure. If subsequent work is short, or if most observations remain relevant, the overhead of extra prompts, notes, and tool calls can outweigh the reduction. Planroom provides a concrete boundary case.

Working with verbose external tools. The retention operation acts on recorded observations, so an upstream tool need not expose its own compaction controls. In principle, the same contract can cover documentation, diagnostic logs, browser text, and images. Its value depends on whether the agent can retain the useful evidence after inspection, not merely on the size of a response. Filtering and curation can be combined: one limits initial exposure, while the other revisits retention as the task evolves.

Task pivots. A new user request supplies information about what should remain active. For example, after a diagnosis is complete, the agent can retain the finding and its source references while archiving the long observations used to establish it. Batch selection permits a separate note

for each result, preserving the surrounding conversation and access to the originals rather than replacing the entire session with a single summary. The same-session continuation exercises the mechanism beyond one isolated answer. It shows that the method can continue implementing after context edits while retained history continues to grow. An identical-history branching comparison is the next step for isolating the effect of curation at the pivot itself.

Why retained history first, and context editing next. Retained history answers the initial deployment question: what changes when an acting agent gains explicit control over which tool results remain active? It measures the combined interface and prompting policy against keeping all observations. The next direct comparison should test this policy against threshold-triggered removal of older tool results, as documented in Anthropic’s context-editing documentation (Anthropic, n.d.). That comparison would ask whether task-aware selection, per-result notes, and recovery improve the resource–quality trade-off beyond automatic clearing. A controlled test should use the same acting model, tools, tasks, and output budget across conditions; a harness implementation of an analogous clearing policy must be distinguished from a test of the provider’s actual feature. This baseline is a priority for the next study and has not been evaluated in the present experiments.

Cache and control. The cost result is compatible with reduced input outweighing cache disruption, but it does not identify the provider’s cache mechanics. More model decisions can coexist with lower total cost, as in OpenTelemetry, or with a small cost increase, as in Planroom. Curation frequency is consequently not a useful success metric on its own. ACON includes acting-agent and compressor calls in its practical API cost estimate but explicitly excludes input caching (Kang et al. 2025). Our accounting includes reported cache usage; the different workloads and models preclude a direct numerical comparison.

Protected instructions and remaining risks. Protecting user messages prevents direct archival of the task text. It does not ensure that the agent follows those instructions or preserves the evidence needed to satisfy them. Notes can be incomplete, mistaken, or overly confident; the model may fail to notice when recovery is necessary. Exact archive integrity addresses only one part of that problem.

7 Limitations and next experiments

The principal limitation is sampling: one OpenTelemetry pair and one Planroom pair do not estimate the variance of the policy’s effect. The Sphinx batch has three sessions per arm but only one authored task and an older configuration. There is no preregistered held-out task distribution, no randomized OpenTelemetry order, and no evidence for a universal degradation threshold. Confidence intervals over requests or individual grader cases would incorrectly treat dependent observations as independent replicates.

The method condition includes instructions, examples, reminders, and tools. No ablation separates these components, and no comparison tests provider-side context editing, a learned manager, chronological eviction, summarization, or programmatic filtering. The experiments therefore establish neither superiority over other context-management systems nor novelty of the shared archival and recovery design. Task trajectories also differ: retained OpenTelemetry has 275 logical image occurrences in requests while method has none. These are repeated input occurrences, not 275 unique screenshots. Provider usage is aggregate and does not expose a separate image-cost decomposition. We cannot attribute the entire between-arm resource difference to text archival alone.

The archive audit establishes byte-exact preservation, not the sufficiency of every note or the dispensability of every removed result. With no recovery in the main pair, its effect on model

decisions and the reliability of downstream retrieval remain unmeasured. The browser continuation also has different inherited histories and unblinded coarse grading. Smaller active context alone is not a measure of greater reasoning ability or a guarantee of preserved quality.

Quality measurement is uneven. The primary oracle covers two specific behaviors; the currency score combines dynamic checks with syntax-sensitive static heuristics; Planroom includes trace-based assessment; and the browser continuation is manually scored. Shared checks and requirements make individual grader cases dependent; passing the two-case primary oracle does not establish complete task correctness or statistical equivalence. Remaining-budget code in both follow-up solutions does not by itself establish a correct end-to-end timeout bound. Gold/baseline separation validates the instrument against its reference candidates but does not eliminate these limits. Before interpreting follow-up score differences as capability improvements, broader implementation-agnostic behavioral tests are needed.

The system is evaluated through one provider model identifier. Provider-side checkpoint identity, inference hardware, cache internals, energy use, and training-data overlap with public repositories are unavailable. Run-start configuration and tariffs are recorded, but exact model and environment determinism is not guaranteed. The hosted inference figures exclude local build, browser, evaluation, and researcher labor costs.

The next study should freeze the corrected setup, run balanced repeated sessions with randomized order, and define task-level outcomes before inspection. An identical-history pivot experiment would isolate retention changes from earlier trajectory differences. A recovery-demand task should make previously archived details necessary and measure retrieval success and downstream correctness. Finally, direct comparisons against compacting and filtering baselines, including ACON and ACM, are required to evaluate the practical value of the specific interface, rather than the value of context reduction in general.

8 Conclusion

Agent-controlled forgetting provides an explicit way for a tool-using model to distinguish information it needed to inspect from information it still needs to carry. In the central noisy debugging case, reversible curation accompanies a 75% smaller final measured prompt, 50% less cumulative input, and substantially lower estimated API cost, while both arms pass the same two primary behavioral cases. The method also requires more actions and time, and follow-up correctness remains incomplete. Contrasting application development and an adverse continuation show that neither resource savings nor quality preservation follows automatically from a smaller active history. The practical result is that an agent can complete noisy tool-driven work with substantially less active context and repeated input, using explicit archival decisions during the task. The contrasting outcomes define where further comparisons and replication are needed.

9 Reproducibility, provenance, and disclosure

The local workspace preserves prompts, task snapshots, request traces, archives, submitted workspaces, tariff ledgers, and independent grading artifacts. The OpenTelemetry manifest records the frozen benchmark and input hashes. The evidence map specifies the artifact behind each central claim. The author is Jan-Peter Franke, based in Münster, Germany. The companion research artifacts are available in the [public repository](#). It includes the main-case harness and benchmark inputs, submitted candidates, and numerical evidence for the reported comparisons. Full provider transcripts and third-party literature copies are not part of that release package. Author-owned research code is released under the MIT license and derived numerical records

under CC BY 4.0; third-party code retains its original licenses. The paper uses the arXiv perpetual non-exclusive distribution license.

The harness and this manuscript were developed with AI assistance under the author’s direction. AI assistance was also used in experiment construction and analysis, so author review of interpretation and evaluation code is required before submission. No model training was performed for this study. Archival preserves information rather than erasing it, and a deployed system must account for that persistence when handling sensitive inputs. We do not infer energy savings from token or price reductions.

References

- Anthropic. n.d. *Context Editing*. <https://platform.claude.com/docs/en/build-with-claude/context-editing>.
- Conner, Shane. n.d. *Pi-Fold: Lossless Context Folding for the Pi Coding Agent*. <https://github.com/shaneconner/fold/blob/7eb2ab47e291f4616ba4850e3648724fa5600a16/README.md>.
- Kang, Minki, Wei-Ning Chen, Dongge Han, et al. 2025. *ACON: Optimizing Context Compression for Long-Horizon LLM Agents*. <https://arxiv.org/abs/2510.00615>.
- Li, Xiaochuan, Ryan Ming, Meng Chu, Shuai Shao, Rong Jin, and Chenyan Xiong. 2026. *ACM: Agentic Context Management for Long Horizon Tasks*. <https://arxiv.org/abs/2607.23809>.
- Liu, Nelson F., Kevin Lin, John Hewitt, et al. 2024. “Lost in the Middle: How Language Models Use Long Contexts.” *Transactions of the Association for Computational Linguistics* 12: 157–73. https://doi.org/10.1162/tacl_a_00638.
- Nahid, Md Mahadi Hasan, and Davood Rafiei. 2025. *PRISM: Agentic Retrieval with LLMs for Multi-Hop Question Answering*. <https://arxiv.org/abs/2510.14278>.
- Packer, Charles, Sarah Wooders, Kevin Lin, et al. 2023. *MemGPT: Towards LLMs as Operating Systems*. <https://arxiv.org/abs/2310.08560>.
- Rusu, Theo, Sourena Khanzadeh, and Manar Alalfi. 2026. *Selective Forgetting: A Graph-Based Memory Framework for Long-Term LLM Agents*. <https://arxiv.org/abs/2608.28978>.
- Sun, Weiwei, Miao Lu, Zhan Ling, et al. 2025. *Scaling Long-Horizon LLM Agent via Context-Folding*. <https://arxiv.org/abs/2510.11967>.
- Wei, Lei, Xiao Peng, Xu Dong, Niantao Xie, and Bin Wang. 2026. *FadeMem: Biologically-Inspired Forgetting for Efficient Agent Memory*. <https://arxiv.org/abs/2601.18642>.
- Ye, Rui, Zhongwang Zhang, Kuan Li, et al. 2025. *AgentFold: Long-Horizon Web Agents with Proactive Context Management*. <https://arxiv.org/abs/2510.24699>.
- Yi, Lu, Runlin Lei, Liuyi Yao, et al. 2026. *Learning Agent-Compatible Context Management for Long-Horizon Tasks*. <https://arxiv.org/abs/2605.30785>.
- Zhang, Qizheng, Changran Hu, Shubhangi Upasani, et al. 2025. *Agentic Context Engineering: Evolving Contexts for Self-Improving Language Models*. <https://arxiv.org/abs/2510.04618>.
- Zhang, Yuxiang, Jiangming Shu, Ye Ma, Xueyuan Lin, Shangxi Wu, and Jitao Sang. 2026. “Memory as Action: Autonomous Context Curation for Long-Horizon Agentic Tasks.” *Findings of the Association for Computational Linguistics: ACL 2026*, 19149–64. <https://doi.org/10.18653/v1/2026.findings-acl.956>.